

CHAPTER 1

1.1 INTRODUCTION

Scuba diving, with its mesmerizing underwater landscapes and the thrill of exploring the ocean's depths. Exploration of the underwater world, whether for scientific research, recreational diving, or industrial purposes, has long been a captivating but challenging endeavour. Ensuring the safety and well-being of scuba divers in the depths of the ocean is of paramount importance. To address this concern, a groundbreaking technology known as Li-Fi-based Scuba Diver's Health Monitoring System has been developed, offering real-time health monitoring and a remarkable feature for handling abnormal health situations. This innovative system leverages Li-Fi, a wireless communication technology that uses light to transmit data. Unlike traditional radio-based communication systems, Li-Fi operates reliably underwater, making it an ideal choice for maintaining a connection between the diver and the surface. The primary objective of this system is to continuously monitor the diver's vital signs, including heart rate, oxygen levels, and body temperature. This real-time health data transmission allows for instant awareness of the diver's well-being during their underwater expedition. However, the true innovation lies in the system's response to abnormal health situations. In cases where the diver's health parameters deviate from the norm in a potentially dangerous manner, the system triggers a rapid and proactive response. This response takes the form of an air balloon that is carried by the scuba diver. When an abnormal health situation is detected, the air balloon swiftly inflates and rises to the surface. Li-Fi (Light Fidelity) is a wireless communication technology that uses light signals to transmit data. In our system, Li-Fi is employed to enable high-speed data transmission from the diver's equipment to the surface, ensuring rapid updates on the diver's status. This technology is reliable and avoids electromagnetic interference underwater. This ingenious "popout" mechanism ensures the diver's safe ascent, avoiding potential risks associated with underwater emergencies. Simultaneously, the system notifies the receivers on the surface of the situation. The surface receivers are equipped to provide immediate alerts, enabling a swift response to the diver's condition. This dual approach - immediate ascent and surface-side notification significantly enhances the safety of scuba divers, reducing the risk of injury or fatality in challenging underwater environments. To further enhance safety, a Panic Switch is integrated into the system.

1.2 OBJECTIVES

- **Real-Time Health Monitoring:**
Continuously monitor vital signs such as heart rate, oxygen levels, and body temperature of scuba divers using wearable sensors.
- **Reliable Underwater Communication:**
Ensure accurate and real-time transmission of health data to the surface using Li-Fi technology, even in harsh underwater environments.
- **Abnormal Health Detection:**
Implement algorithms to detect critical deviations in health parameters and trigger immediate alerts.
- **Emergency Ascent System:**
Design an air balloon-based emergency ascent mechanism for automatic surfacing when a diver's health is at risk.
- **Surface Alert System:**
Enable real-time surface-side notifications to allow prompt response and support from the monitoring team.

1.3 PROBLEM STATEMENT

Scuba divers face significant risks during underwater missions due to the lack of effective communication systems and continuous health monitoring capabilities. Traditional wireless technologies like radio frequency and Wi-Fi fail to function efficiently underwater, making it difficult to transmit critical information such as vital health data or emergency alerts. This communication gap increases the danger of undetected health issues, delayed rescue operations, and overall operational inefficiency. Therefore, there is a need for a reliable, real-time solution that can monitor divers' health conditions and facilitate seamless underwater communication. This project addresses this need by implementing Li-Fi technology to enable light-based communication and real-time health monitoring for enhanced diver safety and mission effectiveness.

1.4 EXISTING SYSTEM

Underwater wireless communication has always posed significant technical challenges due to the unique physical properties of water, which strongly attenuate conventional radio frequency (RF) signals. As a result, early underwater communication systems relied on alternative transmission methods such as infrared (IR) radiation. In a typical infrared-based underwater wireless communication system, communication is achieved using two primary modules: a transmitter module and a receiver module, together commonly referred to as a transceiver system. The transmitter converts electrical data signals into infrared light signals, which propagate through the water medium, while the receiver detects the incoming infrared light and converts it back into electrical signals for further processing.

In this system, the transmitter generally consists of an infrared LED or laser diode that emits IR radiation when driven by an electrical signal. The receiver module typically employs a photodiode or phototransistor that is sensitive to infrared wavelengths. These components enable short-range point-to-point communication between underwater devices. Due to the directional nature of infrared radiation, such systems often require accurate alignment between the transmitter and receiver to ensure reliable data transmission. The use of infrared communication provides certain advantages, including immunity to electromagnetic interference and reduced signal leakage compared to RF-based systems.

However, despite these benefits, infrared-based underwater communication systems exhibit several critical limitations that restrict their effectiveness and scalability. One of the major drawbacks is the limited transmission range, as infrared signals suffer from significant absorption and scattering in water, especially in turbid or polluted environments. Particles, suspended matter, and water salinity further degrade the signal strength, leading to frequent data loss. Additionally, these systems require a clear line-of-sight (LOS) between the transmitter and receiver, making them highly sensitive to misalignment caused by water currents or diver movement.

CHAPTER 2

LITERATURE SURVEY

[1] Gunanandhini S. et al.

Gunanandhini S. *et al.* proposed an underwater wireless communication system for monitoring the health of scuba divers. The system continuously tracks physiological parameters such as blood pressure, heart rate, and risk assessment indicators. It enables real-time transmission of health data from the underwater environment to the nearest ship, ensuring immediate response during emergencies. The use of wireless communication enhances diver safety by providing continuous health surveillance even in challenging underwater conditions.

[2] Sasikala S. et al.

Sasikala S. *et al.* focused on a Li-Fi-based health monitoring system for scuba divers. The system monitors vital parameters including heartbeat, temperature, and respiratory organ expansion. Health data is stored in a microcontroller for further analysis and transmitted only during abnormal conditions to reduce power consumption. The proposed system supports operation up to 135 meters underwater and allows data transmission to multiple receivers simultaneously, making it suitable for deep-sea diving applications.

[3] A. Senthilkumar et al.

A. Senthilkumar *et al.* presented a smart lifesaver system designed to monitor the health of scuba divers using underwater data transmission techniques. The study highlights the potential of underwater communication technologies for reliable data exchange. The authors emphasize the application of underwater vehicles, gliders, and moorings for collecting spatial and temporal data, extending these concepts to improve diver safety and real-time health monitoring.

[4] Vivek Sangani et al.

Vivek Sangani *et al.* proposed a Li-Fi-based patient monitoring system aimed at reducing radio-frequency interference with the human body. The system uses sensors such as temperature, heartbeat, and motion sensors to collect physiological data. An ATmega8 microcontroller processes the data and converts it into optical signals transmitted through LEDs or lasers. The received signals are decoded using photodetectors and displayed on a computer or monitoring unit, demonstrating the effectiveness of Li-Fi for health monitoring applications.

[5] Subodh Bhosale et al.

Subodh Bhosale *et al.* developed a novel diver health monitoring system for continuous, cuffless blood pressure measurement using pulse arrival time (PAT). The system analyzes ECG and PPG signals to estimate blood pressure accurately. Experimental validation showed a strong correlation between PAT and blood pressure levels. The proposed approach enables early detection of sudden blood pressure variations and provides timely alerts, ensuring enhanced diver safety through non-invasive monitoring.

[6] Balaji K. et al.

Balaji K. *et al.* investigated the integration of IoT with underwater communication using Li-Fi technology. The study addressed challenges related to underwater optical communication and acoustic limitations. The system transmits received optical signals to a sink node, which forwards the data to the cloud. The information can be accessed via a mobile application, highlighting the feasibility of Li-Fi-based IoT systems for underwater data transmission and remote monitoring.

[7] Porselvi S. et al.

Porselvi S. *et al.* proposed a healthcare monitoring system using Li-Fi networks within hospital environments. The system monitors vital parameters such as temperature, pressure, heartbeat, glucose level, and respiratory conditions. Sensor data is processed using a PIC microcontroller and displayed graphically. A prototype based on visible light communication was developed and tested, demonstrating reliable and secure medical data transmission using Li-Fi.

[8] Kusanur et al.

Kusanur *et al.* evaluated the performance of Li-Fi-based underwater communication systems under varying water quality conditions. Experimental analysis using LED transmitters and photodiode receivers measured signal attenuation and data rate performance in different turbidity and salinity levels. The results confirmed the suitability of visible light communication with optimized modulation techniques for real-time underwater data transmission.

[9] Domingos et al.

Domingos *et al.* conducted a comprehensive analysis of modern underwater communication systems to evaluate their effectiveness for real-time monitoring applications. The study involved an extensive literature survey and system-level evaluation of underwater communication techniques. The authors proposed a hybrid architecture combining optical and acoustic communication models to enhance reliability and adaptability in diverse underwater environments.

[10] Kaushal H. et al.

Kaushal H. *et al.* presented a comprehensive study on underwater optical wireless communication (UOWC) systems and their applicability in real-time monitoring applications. The authors analyzed light propagation characteristics in underwater environments, considering absorption, scattering, and turbulence effects. Their study highlighted that blue-green wavelength bands offer superior transmission efficiency underwater, making optical communication suitable for high-data-rate health monitoring systems for scuba divers.

[11] Akyildiz I. F. et al.

Akyildiz I. F. *et al.* proposed an underwater wireless sensor network (UWSN) architecture for environmental and human monitoring applications. The system integrates sensor nodes capable of monitoring physiological and environmental parameters and transmitting data to surface stations. The authors emphasized challenges such as limited bandwidth, high propagation delay, and energy constraints, and suggested hybrid acoustic–optical communication models to improve reliability for underwater health monitoring.

[12] Zeng Z. et al.

Zeng Z. *et al.* conducted an in-depth survey on underwater optical wireless communication technologies. The study compared optical, acoustic, and radio-frequency underwater communication techniques. The authors concluded that optical communication provides low latency and high data rates over short distances, making it ideal for applications such as diver health monitoring, where real-time transmission of vital signs is critical.

CHAPTER 3

METHODOLOGY

The proposed project aims to develop an innovative wearable health monitoring system for scuba divers, specifically engineered to ensure diver safety during underwater activities. The system integrates a suite of biomedical sensors that continuously measure and monitor key physiological parameters such as heart rate, blood oxygen saturation (SpO₂), and body temperature. These vital signs are crucial indicators of a diver's health status, especially in extreme underwater environments, where conditions such as hypoxia, hypothermia, or cardiac stress can arise unexpectedly.

At the core of the system is a microcontroller, which acts as the processing unit. It collects real-time data from the sensors, filters and analyzes the signals, and interprets the diver's physiological state. To enable real-time communication with the surface team, the system employs Li-Fi (Light Fidelity) technology, which uses modulated LED light to transmit data wirelessly through water. Unlike conventional wireless methods such as radio frequency (RF), which are unreliable underwater, Li-Fi offers a robust and efficient medium for short-range data transmission in aquatic environments, making it ideal for scuba diving applications.

A critical feature of the system is its built-in decision-making algorithm. This algorithm continuously evaluates the incoming health data against predefined safety thresholds. If an abnormal condition is detected—such as dangerously low oxygen levels or irregular heart rate—the algorithm triggers a fail-safe mechanism. This includes the activation of an emergency air balloon system, which helps the diver ascend gently and safely to the water surface. This automated response is designed to minimize the risk of unconsciousness or injury due to delayed human reaction during emergencies.

Simultaneously, the system sends real-time alerts to the surface monitoring station via the same Li-Fi communication channel. These alerts include detailed diagnostic information that allows surface personnel to prepare and initiate appropriate emergency protocols.

3.1 System Architecture Overview

- Clear separation between transmitter and receiver modules
- Sensor integration details
- Communication flow

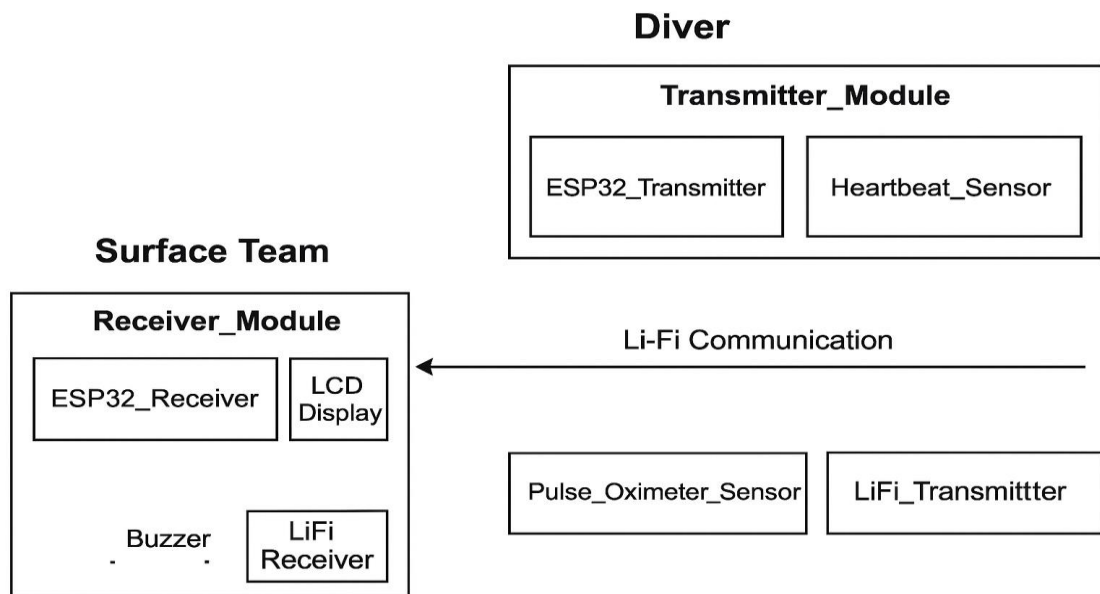


Fig 3.1 : Over view of the project

This system architecture is designed to monitor the health of a scuba diver underwater and transmit the data to the surface team using Li-Fi communication.

The Diver Unit (Transmitter Module) consists of a heartbeat sensor / pulse oximeter sensor and an ESP32 transmitter. The sensors continuously measure vital health parameters such as heart rate and oxygen level. The ESP32 processes this data and sends it through a Li-Fi transmitter in the form of light signals.

The Li-Fi communication link is used to transfer data from the diver to the surface. Li-Fi is suitable for underwater communication because radio waves do not work effectively underwater, while visible light provides better short-range data transmission.

The Surface Team Unit (Receiver Module) includes a Li-Fi receiver, ESP32 receiver, LCD display, and a buzzer. The Li-Fi receiver captures the transmitted light signals and converts them into electrical signals. The ESP32 decodes the data and displays the diver's health

information on the LCD. If abnormal readings are detected, the buzzer is activated to alert the surface team.

3.1.1 BLOCK DIAGRAM OF AN RECEIVER MODULE

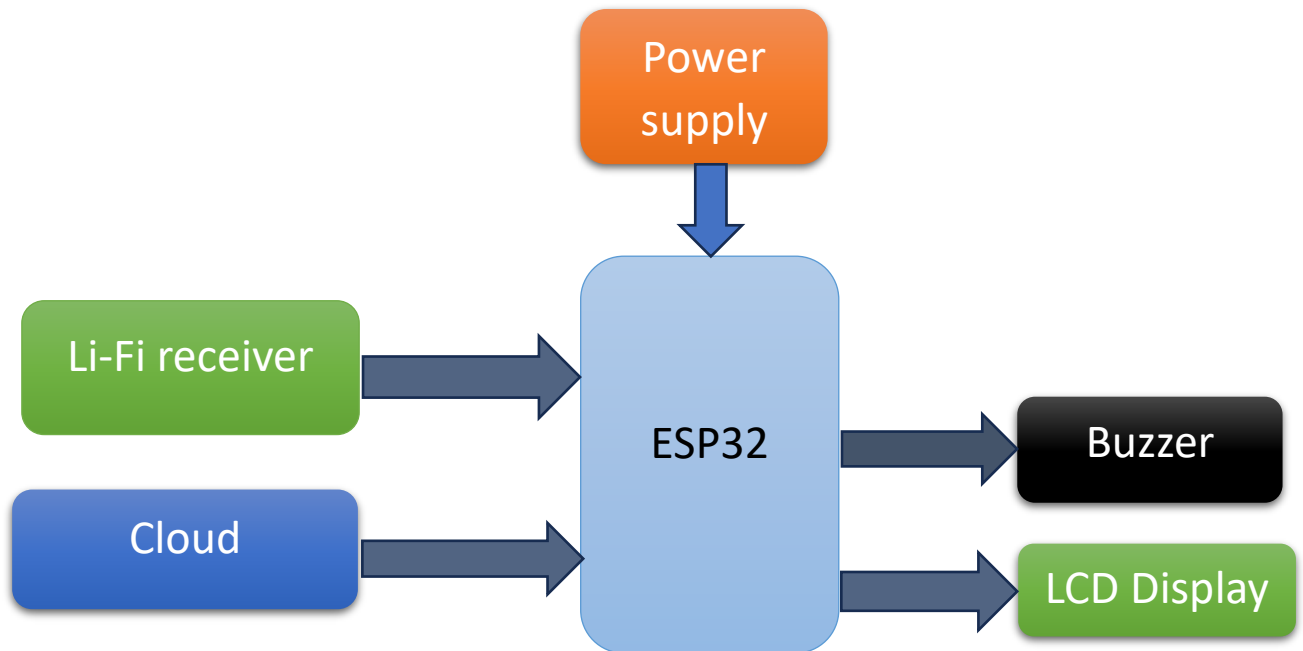


Fig 3.2 Receiver module block diagram

Surface Team Unit (Receiver Module)

The Surface Team Unit receives, processes, and displays the diver's health data.

- **Li-Fi Receiver**
A photodiode-based Li-Fi receiver captures the optical signals transmitted by the diver unit and converts them back into electrical signals.
- **ESP32 Receiver**
The ESP32 decodes the received data and processes it for visualization and alert generation. It ensures accurate interpretation of the diver's health status.
- **LCD Display**
The LCD display provides real-time visualization of vital parameters such as heart rate and oxygen levels, allowing the surface team to continuously monitor the diver's condition.
- **Buzzer (Alert System)**
The buzzer acts as an emergency alert mechanism. It is activated when abnormal health conditions are detected, such as dangerously high or low heart rate or oxygen levels, ensuring immediate attention and response from the surface team.

3.1.2 BLOCK DIAGRAM OF TRANSMITTER MODULE

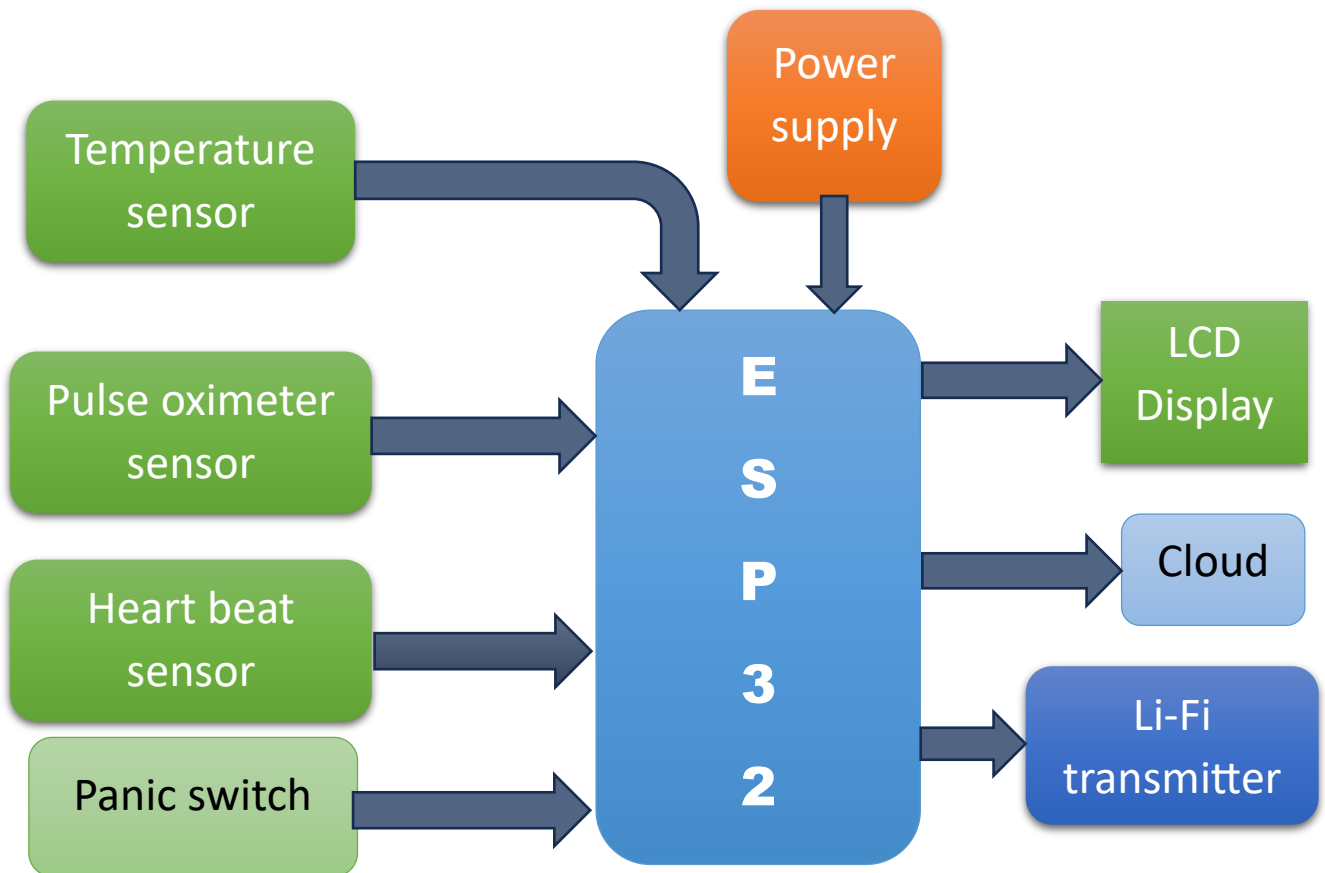


Fig 3.3 Transmitter module block diagram.

Diver Unit (Transmitter Module)

The Diver Unit is worn by the scuba diver and is responsible for sensing and transmitting physiological data.

- Heartbeat Sensor / Pulse Oximeter Sensor

These sensors continuously measure vital parameters such as heart rate and blood oxygen saturation (SpO₂). Monitoring these parameters is critical for detecting early signs of stress, fatigue, or medical emergencies such as hypoxia.

- ESP32 Transmitter

The ESP32 microcontroller acts as the core processing unit. It acquires sensor data, performs basic signal conditioning, and prepares the data for transmission. The ESP32 is chosen due to its low power consumption, fast processing capability, and ease of integration with sensors.

- **Li-Fi Transmitter**

The processed health data is converted into optical signals and transmitted using a Li-Fi transmitter (LED or laser source). Li-Fi is preferred over RF communication because radio waves attenuate rapidly underwater, whereas visible light can provide high-speed and low-latency communication over short distances.

3.1.3 Microcontroller ESP 32 Transmitter and Receiver Module.



Fig 3.4 ESP 32

The ESP32 is a powerful and energy-efficient microcontroller developed by Espressif Systems, widely used in IoT, automation, and embedded system applications. It features a dual-core Tensilica LX6 processor, integrated Wi-Fi and Bluetooth (Classic + BLE), and supports a wide range of peripherals, making it highly versatile for both simple and advanced projects. The ESP32 includes built-in modules such as ADC, DAC, PWM, Timers, SPI, I2C, UART, CAN, and Touch Sensors, enabling seamless interfacing with various sensors and actuators. It also supports deep sleep modes, making it ideal for battery-powered systems. With its strong processing capability, low power consumption, rich communication features, and compatibility with Arduino IDE, Micro Python, and ESP-IDF, the ESP32 has become one of the most popular choices for modern IoT and embedded applications.

Components of an ESP 32 module

1. ESP32 Dual-Core Microcontroller (Xtensa LX6)

The heart of the ESP32 board is its powerful dual-core Xtensa LX6 processor, capable of running at 160–240 MHz. It manages all computations, sensor processing, communication tasks, and program execution.

2. Wi-Fi Module (802.11 b/g/n)

This built-in wireless module enables the ESP32 to connect to Wi-Fi networks for IoT applications. It supports both Station and Access Point modes, allowing devices to communicate over the internet or local networks.

3. Bluetooth Module (Bluetooth Classic + BLE)

ESP32 includes both Bluetooth Classic and Bluetooth Low Energy (BLE). This allows low-power data transfer, device pairing, mobile communication, and wireless control.

4. Flash Memory (Typically 4–16 MB)

The flash memory stores the program code, Wi-Fi credentials, libraries, and other important data. It ensures the ESP32 retains the program even after powering off.

5. GPIO Pins (General Purpose Input/Output)

ESP32 provides 30+ GPIO pins used for connecting sensors, modules, LEDs, motors, displays, etc. These pins support PWM, ADC, DAC, UART, SPI, I2C, and more.

6. ADC (Analog-to-Digital Converters)

The ESP32 has 12-bit ADC channels used to measure analog signals such as temperature, light intensity, gas sensors, and other variable voltages.

7. DAC (Digital-to-Analog Converters)

It includes two 8-bit DAC channels that convert digital values into analog signals, useful for audio output and analog control applications.

8. Power Management IC (PMIC)

This chip regulates all power supplied to the ESP32 module. It allows the board to run from 3.3V input, manages sleep modes, and supports low-power operations to save battery.

9. Internal Clock and Crystal Oscillator

A 40 MHz crystal oscillator ensures stable timing for Wi-Fi, Bluetooth, and the microcontroller's internal operations.

10. PCB Antenna / External Antenna Connector

ESP32 boards come with an onboard chip antenna or an external antenna connector, enabling strong and stable wireless signal transmission and reception.

11. USB-to-UART Bridge (CP2102/CH340)

Most development boards include a USB-to-Serial converter allowing you to upload programs from your computer through the micro-USB port.

12. Reset & Boot Buttons

- **Reset button:** Restarts the microcontroller.
- **Boot/Flash button:** Used during firmware upload or entering flashing mode.

13. Voltage Regulator (3.3V)

The regulator converts 5V from USB to 3.3V, which the ESP32 chip requires to operate safely.

3.1.4 LI-FI Transmitter Module

Working of LI-FI :

Li-Fi is typically implemented using white LED light bulbs at the downlink transmitter. These devices are normally used for illumination only by applying a constant current. However, by fast and subtle variations of the current, the optical output can be made to vary at extremely high speeds . This very property of optical current is used in Li-Fi setup. The operational procedure is very simple-, if the LED is on, you transmit a digital 1, if it's off you transmit a 0. The LEDs can be switched on and off very quickly, which gives nice opportunities for transmitting data.

Li-Fi, or "light fidelity", is a technology, that can be a complement of RF communication (Wi-Fi or Cellular network), or a replacement in contexts of data broadcasting. Li-Fi, like Wi-Fi, is the high speed, bidirectional and fully networked subset of visible light communications (VLC). It is wireless and uses visible light communication (instead of radio frequency waves), which carries much more information, and has been proposed as a solution to the RF-bandwidth limitations.

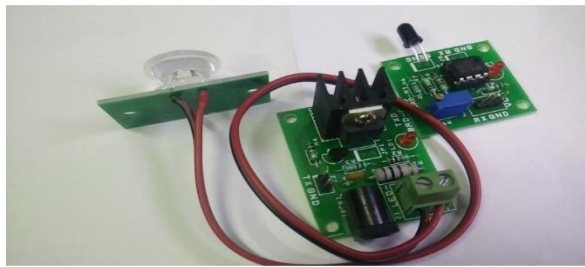


Fig 3.5 LI-FI Transmitter

Simple module of Li-Fi Tx which have two stages pre-amplifier and power amplifier. At the pre- amplifier circuit in U1 picks the transducer output, which catches the very low pitch sound waves. In pre-amplifier amplifies voltage sufficiently the condenser microphone which is biased through R1 changes its internal-resistance with respect to the picked sound waves. This varying signal drives the U1 IC LM 741 through coupling capacitor C1. Hence the corresponding amplified output signals are observed at terminal 6 of IC U1& Through P1 we are controlling the gain. The input signals from condenser microphone and output signals at terminal 6 are coupled by capacitors C2 to power amplifier input terminal 3 of IC U2. In power amplifier stage. Amplifies both current and voltage sufficiently then transmitted through light Transmission.

MICROCONTROLLER:

A microcontroller is often small and low cost. The components are chosen to minimize size and to be as inexpensive as possible. A microcontroller is often, but not always, ruggedized in some way. The microcontroller controlling a car's engine, for example, has to work in temperature extremes that a normal computer generally cannot handle. A car's microcontroller in Kashmir regions has to work fine in -30degree F (-34° C) weather, while the same microcontroller in Gujarat region might be operating at 120 degrees F (49° C). When you add the heat naturally generated by the engine, the temperature can go as high as 150° or 180° degrees F (65-80° C) in the engine compartment. On the other hand, a microcontroller embedded inside a VCR hasn't been ruggedized at all.

3.1.5 LI-FI Receiver Module

SOLAR



Fig 3.6 LI-FI Receiver

As shown in Fig 3.5 simple module of Li-Fi Rx which have two stages pre-amplifier and power amplifier. At the pre- amplifier stage U1 picks the transmitted signal through LDR RX, which catches the very low pitch sound waves. In pre-amplifier amplifies the voltage sufficiently the light Rx Transistor (LDR) which is biased through R7 & voltage divider circuit is formed using R1 & R8 resistors. The varying signal drives the U1 ic LM 358 through coupling capacitor C2. Hence the corresponding amplified output signals are observed at terminal 7 of IC U1.

LDR:

When it's light, the LDR has low resistance. So the voltage at the base of the transistor is not high enough to turn the transistor on. Therefore, no current will go from the collector to the emitter of the transistor. All the current will instead pass through the potentiometer and the LDR.

A solar cell, or photovoltaic cell, is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect, which is a physical and chemical phenomenon. It is a form of photoelectric cell, defined as a device whose electrical characteristics, such as current, voltage, or resistance, vary when exposed to light. Individual solar cell devices can be combined to form modules, otherwise known as solar panels. The common single junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 to 0.6 volts.

Solar cells are described as being photovoltaic, irrespective of whether the source is sunlight or an artificial light. In addition to producing energy, they can be used as a photodetector (for example infrared detectors), detecting light or other electromagnetic radiation near the visible range, or measuring light intensity.

The operation of a photovoltaic (PV) cell requires three basic attributes:

- The absorption of light, generating either electron-hole pairs or excitons.
- The separation of charge carriers of opposite types.
- The separate extraction of those carriers to an external circuit.

In contrast, a solar thermal collector supplies heat by absorbing sunlight, for the purpose of either direct heating or indirect electrical power generation from heat. A "photo electrolytic cell" (photoelectrochemical cell), on the other hand, refers either to a type of photovoltaic cell (like that developed by Edmond Becquerel and modern dye-sensitized solar cells), or to a device that splits water directly into hydrogen and oxygen using only solar illumination. The term solar panel is used colloquially for a photo-voltaic (PV) module.

3.1.6 Communication System

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A Software Serial library allows for serial communication on any of the microcontroller's digital pins.

The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus, see the documentation for details. For SPI communication, the SPI library is used.

3.1.7 OPTIMIZATION GOALS



Fig 3.7 SCUBA DIVER

1. Primary Goals

- Develop a reliable real-time health monitoring system for scuba divers
- Implement Li-Fi based communication for underwater data transmission
- Create an early warning system for health abnormalities
- Establish instant communication between divers and surface support

2. Technical Goals

- Achieve high-speed data transmission using Li-Fi technology
- Implement robust sensor integration for vital sign monitoring
- Develop low-latency communication protocols
- Create energy-efficient monitoring solutions

3. Safety Goals

- Reduce response time in emergency situations
- Improve accuracy of health monitoring
- Enhance diver safety through proactive alerts
- Provide continuous health status updates to surface teams

4. System Performance Goals

- Achieve reliable data transmission up to 30 meters depth
- Maintain continuous monitoring with minimal interruptions

- Ensure battery life sufficient for typical dive durations
- Implement redundant safety mechanisms

3.2 FRAME DESIGN

3.2.1 LED DISPLAY



Fig 3.8 LED DISPLAY

A Liquid Crystal Display (LCD) is a flat-panel display technology widely used in devices like TV monitors, and smart phones. It works by using liquid crystals that control the passage of light from a backlight source. These crystals change alignment when an electric current is applied, allowing them to modulate light and display images. The screen includes polarizing filters, glass substrates, and color filters to produce visible output.

The 44780 standard requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used the LCD will require a total of 7 data lines (3 control lines plus the 4 lines for the data bus). If an 8-bit data bus is used the LCD will require a total of 11 data lines (3 control lines plus the 8 lines for the data bus).

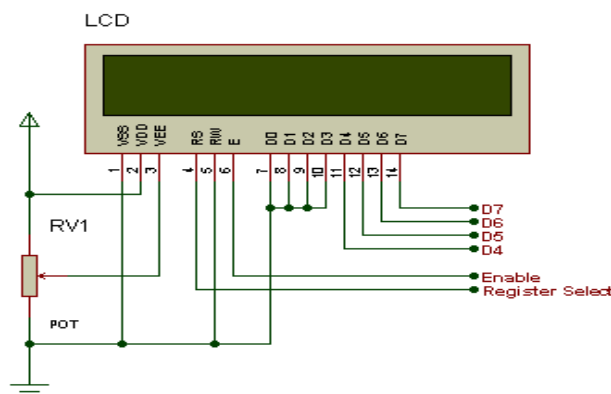


Fig 3.9 Internal structure of LED

The three control lines are referred to as **EN**, **RS**, and **RW**.

The **EN** line is called "Enable." This control line is used to tell the LCD that you are sending it data. To send data to the LCD, your program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring **EN** high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

The **RS** line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which should be displayed on the screen. For example, to display the letter "T" on the screen you would set RS high.

The **RW** line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are write commands--so RW will almost always be low. Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

3.2.1 TEMPERATURE SENSOR



Fig 3.10 Temperature Sensor

A temperature sensor is a device that measures heat to provide temperature readings in the form of electrical signals. Common types include thermocouples, RTDs (Resistance Temperature Detectors), thermistors, semiconductor sensors, and infrared (IR) sensors. Each type has its own strengths: thermocouples handle a wide range, RTDs offer high accuracy, thermistors are

highly sensitive, semiconductor sensors are easy to use with electronics, and IR sensors enable non-contact measurement.

3.2.2 PULSE OXIMETER SENSOR



Fig 3.11 Pulse Oximeter

A pulse oximeter uses light-emitting diodes (LEDs) to shine red and infrared light through the skin. Oxygenated and deoxygenated haemoglobin absorb these lights differently. A photodetector on the opposite side measures the amount of light that passes through, and the device calculates SpO₂ and pulse rate based on the absorption pattern.

3.2.3 HEART BEAT SENSOR

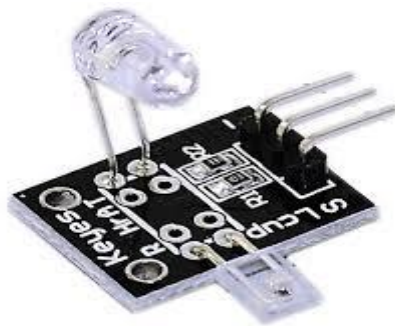


Fig 3.12 Heart beat sensor

A heart beat sensor is a device that detects the heart rate by sensing the pulses of blood flow through the body. It is commonly used in fitness trackers, medical devices, and DIY electronics projects. The sensor typically works using photo plethysmography (PPG), which involves a light source (usually an LED) and a photo detector. The light penetrates the skin, and as the heart beats, the volume of blood changes. These changes affect how much light is absorbed or reflected, and the sensor uses this to calculate the pulse rate.

3.3 MECHANICAL DESIGN DESCRIPTION

RELAY, AIR MOTOR AND BUZZER FOR EMERGENCY DETECTION OF AN SCUBA DIVER.

In a health monitoring system for scuba divers, safety and quick response are the most important goals. To achieve this, components such as a relay, air motor, and buzzer play a crucial role in alerting both the diver and the surface monitoring team during emergency situations. The relay acts as an automatic switching device used to control high-power safety mechanisms like the air motor using low-power signals from the microcontroller. The air motor can be used to assist in buoyancy control, helping lift the diver toward the surface when vital signs such as heart rate, blood oxygen level, or body temperature fall to dangerous levels. This provides an automated rescue response when the diver becomes unconscious or experiences difficulty underwater. The buzzer serves as an immediate audible alarm to warn the diver about abnormal health conditions detected by the sensors. It provides real-time alerts so the diver can take corrective action before the situation becomes critical. Together, the relay, air motor, and buzzer significantly enhance diver safety by providing automatic emergency response, continuous monitoring, and prompt warnings in life-threatening underwater environments.

3.3.1 RELAY FOR CONTROLLING AIR MOTOR



Fig 3.13 RELAY

A relay is an electrically operated switch that uses an electromagnet to mechanically open or close a circuit. It allows a low-power control signal (like from a microcontroller or sensor) to switch a much higher power load safely and effectively. Relays are widely used in automation, industrial equipment, automotive systems, and home appliances to control motors, lights, heaters and other devices.

3.3.1.1 WORKING OF AN RELAY MODULE

A **relay module** is an electrically operated switch that allows a low-power control signal (from a microcontroller like ESP32/Arduino) to switch **high-power devices** such as motors, pumps, and lights. It provides **electrical isolation** between the control side (low voltage) and the load side (high voltage).

1. Input Signal (Control Side)

- A small signal (3.3V or 5V) from the ESP32/Arduino is sent to the relay's **input pin**.
- This activates the **transistor driver circuit** inside the module.

2. Electromagnetic Coil Activation

- The transistor energizes the **relay coil**.
- Once the coil is powered, it generates a magnetic field.

3. Switching Mechanism (Load Side)

- The magnetic field pulls a **metal armature**.
- This action **changes the contact position**:
 - **NO (Normally Open)** contact closes → Load is turned ON
 - **NC (Normally Closed)** contact opens → Load is turned OFF

4. Electrical Isolation

- The control signal and load circuit are separated using an **opto-isolator** in many relay modules.
- This protects the microcontroller from high voltage spikes.

5. Load Activation

- Once the relay switches, it can power external devices like:
 - Motors
 - Air pumps
 - Alarms

- Fans
- Lighting systems

Key Points

- Works like a **remote-controlled electrical switch**.
- Allows low-voltage electronics to safely control high-voltage devices.
- Essential for automation and embedded systems such as home automation, safety devices, and health monitoring modules.

3.3.2 AIR MOTOR FOR LIFTING THE SCUBA DRIVER IN EMERGENCY CONDITION



Fig. 3.14 AIR MOTOR

. An air motor (also called a pneumatic motor) is a device that converts compressed air energy into mechanical rotational motion. It operates by directing pressurized air into a motor, which then drives a shaft to produce torque and rotational speed. Air motors are commonly used in industrial tools, machinery, and equipment where electric motors may not be suitable, such as in explosive or wet environments, because they don't produce sparks and are resistant to harsh conditions.

3.3.3 BUZZER FOR EMERGENCY ALLERT



Fig 3. 15 BUZZER

A buzzer is an audio signalling device that produces sound when an electric current is applied. It is commonly used in alarms, timers, confirmation signals, and electronic projects to provide audible feedback or alerts.

Buzzers come in two main types: active and passive.

An active buzzer has a built-in oscillator, so it produces sound on its own when connected to a power supply—just apply voltage, and it buzzes.

A passive buzzer requires an external signal (like a square wave from a microcontroller) to generate sound.

3.3.3.1 WORKING OF BUZZER

A **buzzer** is an electronic sound-producing device that works by converting electrical energy into sound energy. When a voltage is applied to the buzzer, it creates rapid mechanical vibrations, producing a beep or alarm sound.

1. Electromagnetic Buzzer

- Inside it, there is a **coil**, **magnet**, and a **metal diaphragm**.
- When current flows through the coil:
 - It generates a magnetic field.
 - This magnetic field attracts and releases the metal diaphragm repeatedly.
 - The rapid vibrations of the diaphragm produce **sound waves**.
- These buzzers typically produce a **continuous tone**.

2. Piezoelectric Buzzer

- Contains a **piezoelectric material** (usually ceramic) attached to a metal plate.
- When voltage is applied:
 - The piezoelectric material **expands and contracts** very quickly.
 - This causes the metal plate to vibrate.
 - These vibrations create **audible sound**.
- They are lightweight, low-power, and produce sharp beeps.

Key Points

- Buzzers are used for **alerts, alarms, warnings, notifications**.
- Common in embedded systems, safety devices, timers, and health-monitoring systems.
- They work with both **AC and DC signals**, depending on the type.

3.3.4 PANIC SWITCH FOR EMERGENCY ALERT

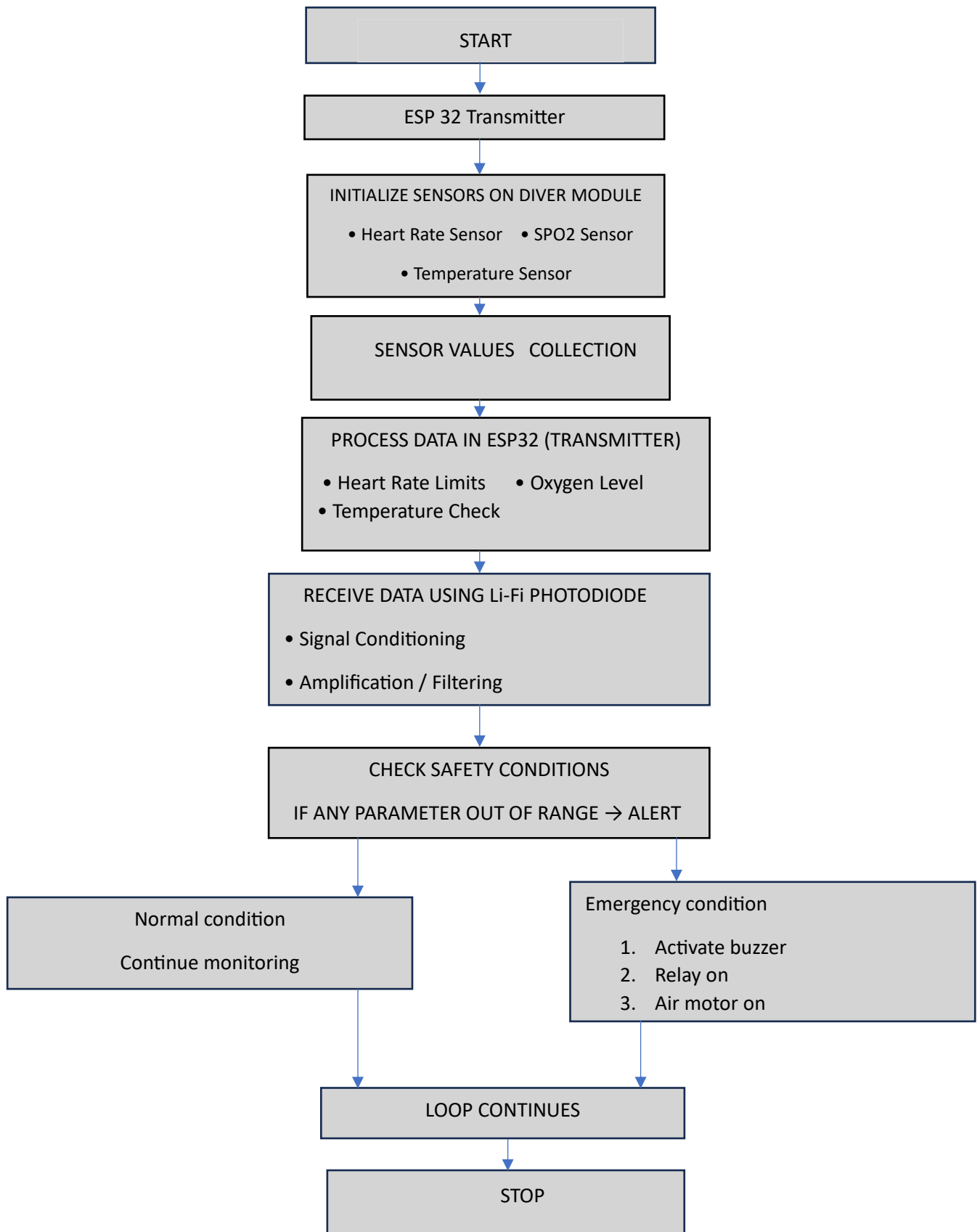


Fig 3.16 Panic switch

The panic switch is a critical safety feature in a scuba diving health monitoring system that allows the diver to manually signal distress to the surface team during emergency situations. Unlike automatic sensor-based alerts, the panic switch provides a direct and immediate way for the diver to request help when they feel unsafe or experience sudden discomfort.

In this system, the panic switch is typically implemented as a waterproof push button integrated into the diver's wearable unit. When pressed, the switch sends a high-priority emergency signal to the ESP32 transmitter, which immediately forwards the alert through the Li-Fi communication link to the surface receiver module.

3.4 FLOW CHART OF THE PROJECT



3.4.1 POWER SUPPLY

The transformer 230Volts will be stepped down to 12-0-12 one side of the 12V is given to the 7805 and LM317. In this project the microcontroller requires +5V power supply. The design description of power supply is given below.

The +5 Volt and 3.8V power supply is based on the commercial 7805 & LM317 voltage regulator IC. This IC contains all the circuitry needed to accept any input voltage from 8 to 18 volts and produce a steady +5 volt & 3.8volt output, accurate to within 5% (0.25 volt). It also contains current-limiting circuitry and thermal overload protection, so that the IC won't be damaged in case of excessive load current; it will reduce its output voltage instead.

The 1000 μ f capacitor serves as a "reservoir" which maintains a reasonable input voltage to the 7805 throughout the entire cycle of the ac line voltage. The bridge rectifier (WM04) keeps recharging the reservoir capacitor on alternate half-cycles of the line voltage, and the capacitor is quite capable of sustaining any reasonable load in between charging pulses.

The LED and its series resistor (220ohm) serve as a pilot light to indicate when the power supply is on and also helps to the reservoir capacitor is completely discharged after power is turned off. Now it's safe to remove or install components for the next experiment

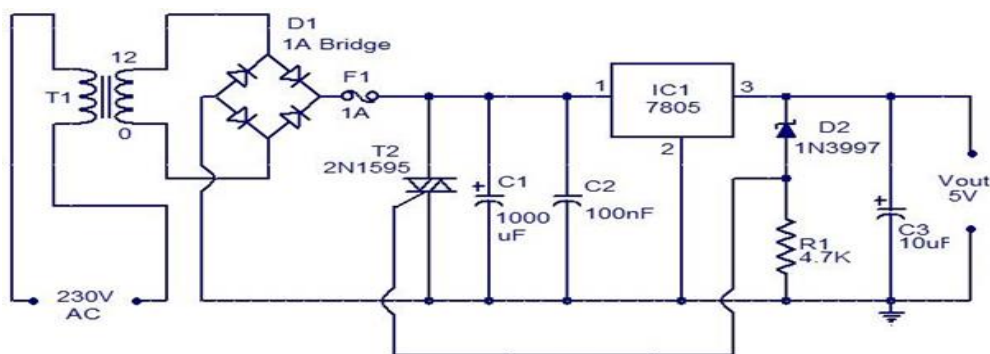


Fig 3.17 Power supply with over voltage protection.

3.5 SOFTWARE DESIGN

INTRODUCTION TO THE ARDUINO IDE

- The Arduino is a single-board microcontroller solution for many DIY projects, we will look at the Integrated Development Environment, or IDE, that is used to program it. Once the installer has downloaded, go ahead and install the IDE. Arduino IDE is an open source software that is mainly used for writing and compiling the code into the Arduino Module.
- It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process.

3.5.1 DOWNLOAD THE IDE

First, you must download the IDE and install it. Start by visiting Arduino's software page. The IDE is available for most common operating systems, including Windows, Mac OS X, and Linux, so be sure to download the correct version for your OS. If you are using Windows 7 or older, do not download the Windows app version, as this requires Windows 8.1 or Windows 10.

3.5.2 The ARDUINO IDE

The Arduino IDE is incredibly minimalistic, yet it provides a near-complete environment for most Arduino-based projects. The middle section of the IDE is a simple text editor that where you can enter the program code. The bottom section of the IDE is dedicated to an output window that is used to see the status of the compilation, how much memory has been used, any errors that were found in the program, and various other useful message

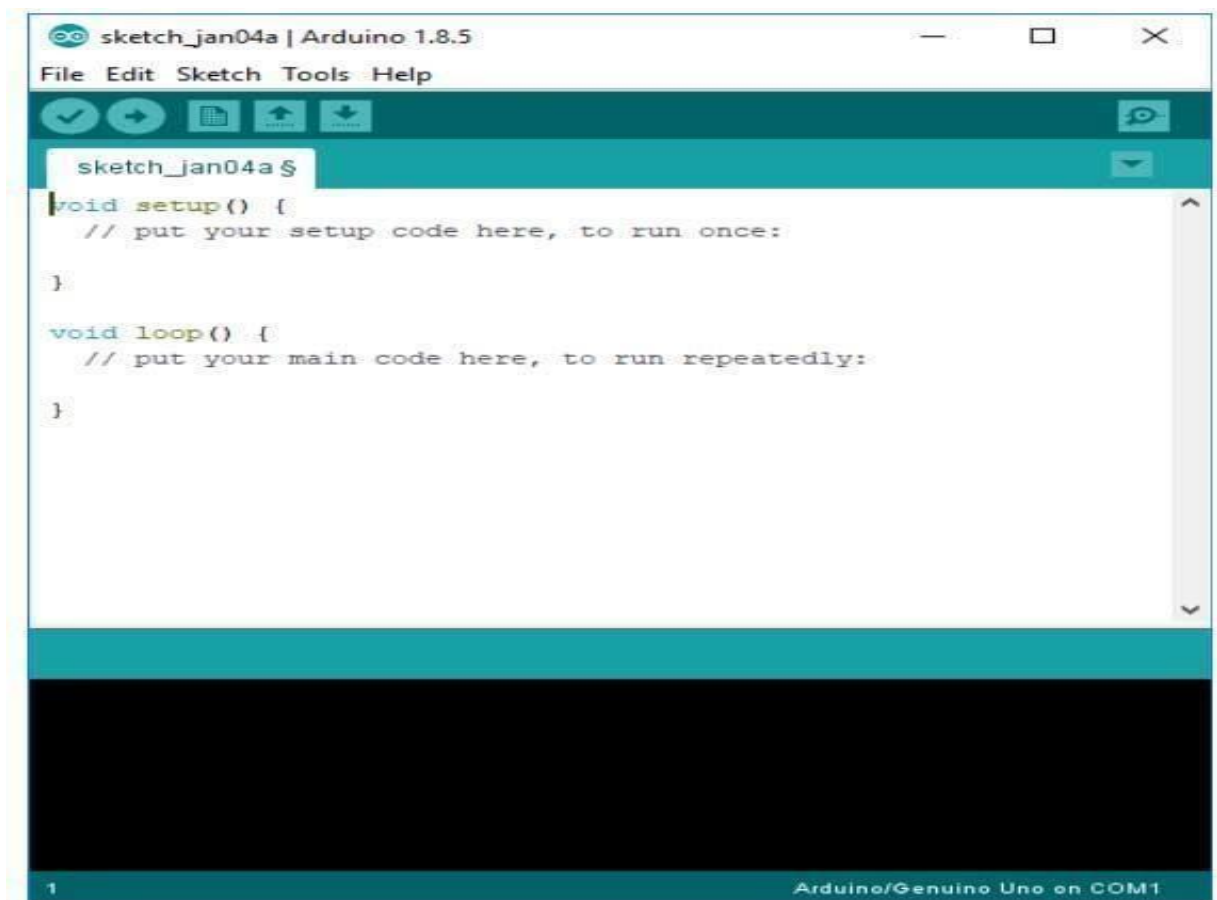


Fig3.18 Arduino program dumping window

Projects made using the Arduino are called sketches, and such sketches are usually written in a cut-down version of C++ (a number of C++ features are not included). Because programming a microcontroller is somewhat different from programming a computer, there are a number of device-specific libraries (e.g., changing pin modes, output data on pins, reading analog values, and timers).

This sometimes confuses users who think Arduino is programmed in an —Arduino language. However, the Arduino is, in fact, programmed in C++. It just uses unique libraries for the device.

THE 6 BUTTONS

While more advanced projects will take advantage of the built-in tools in the IDE, most projects will rely on the six buttons found below the menu bar.



Fig 3.19 The button bar

1. The check mark is used to verify your code. Click this once you have written your code.
2. The arrow uploads your code to the Arduino to run.
3. The dotted paper will create a new file.
4. The upward arrow is used to open an existing Arduino project.
5. The downward arrow is used to save the current file.
6. The far right button is a serial monitor, which is useful for sending data from the Arduino to the PC for debugging purposes.

3.5.4 Functional Requirements

1. The system must monitor vital signs (heart rate, oxygen levels, body temperature) in real-time.
2. The system must transmit health data from the diver to the surface via Li-Fi.
3. The system must detect abnormalities in health parameters and trigger alerts.
4. The system must activate an emergency response in critical situations.
5. The system must display received data on an LCD screen at the surface.
6. The system must include a panic switch for manual distress signalling.
7. The system must provide audible alerts using a buzzer in case of emergencies.

Non-Functional Requirements

1. The system must ensure real-time data transmission with minimal latency.
2. The system must be reliable in underwater environments.

3. The system must have low power consumption to sustain long dives.
4. The system must be lightweight and portable for diver convenience.
5. The system must use secure data transmission to prevent interference.
6. The system must withstand underwater pressure and environmental factors.

3.5.5 Features and Functionalities of Arduino IDE

Programming Language Used

- Arduino IDE uses a simplified version of C/C++.
- It includes built-in functions and libraries to simplify hardware control.
- Programs are written using `setup()` and `loop()` functions.

Compilation and Upload Process

- The IDE converts Arduino code into machine-level instructions using a compiler.
- After successful compilation, the program is uploaded to the microcontroller via USB.
- Errors during compilation are highlighted with line numbers.

Library Management

- Arduino IDE supports a wide range of built-in and external libraries.
- Libraries simplify interfacing with sensors, displays, communication modules, etc.
- The Library Manager allows easy installation and updating of libraries.

Board and Port Selection

- Different Arduino boards (Uno, Nano, Mega, ESP boards, etc.) can be selected.
- Correct COM port selection is required for successful program upload.
- Board settings affect memory usage and pin mapping.

CHAPTER 4

RESULT AND ANALYSIS

The implementation of the scuba diver safety system using Li-Fi technology has demonstrated promising results in ensuring real-time health monitoring and secure underwater communication. The system successfully captured and transmitted the diver's vital signs, including heart rate, oxygen levels, and body temperature, from the underwater module to the surface station. The experimental setup validated the ability of Li-Fi technology to transmit data efficiently in an aquatic environment, reducing the communication gap between divers and the surface team.

One of the key findings of the study was the system's capability to detect abnormal health conditions. When the diver's vital signs deviated from the normal range, the system promptly triggered alerts and notified the surface team. The panic switch functionality was also tested, and it effectively transmitted emergency signals, ensuring a quick response to critical situations. The timely alerts significantly enhance diver safety by allowing for immediate intervention.

The Li-Fi transmission module played a crucial role in ensuring data accuracy and minimizing signal loss underwater. Unlike traditional radio frequency (RF) communication, which faces significant attenuation underwater, Li-Fi technology leveraged light waves for data transmission, achieving greater reliability. However, the system was found to be sensitive to water turbidity and external light sources, which could impact data integrity under certain conditions.

4.1 SENSOR DATA ACQUISITION AND PERFORMANCE

Heart Beat (Pulse Rate)

Body Temperature

SpO2 (Blood Oxygen Level)

Emergency Switch Status

4.1.1 HEARTBEAT SENSOR

The heart beat sensor (MAX30102) performed with high accuracy and stability throughout the testing phase of the system. The sensor produced clean and well-defined photoplethysmography (PPG) waveforms, with distinct systolic and diastolic peaks visible on the monitoring dashboard. During real-time monitoring, the heart rate readings consistently fell within a normal physiological range of 72–110 BPM, and the system reacted instantly to any fluctuations. The graph displayed on the dashboard showed smooth, continuous curves with minimal noise, confirming strong signal quality and proper fingertip placement detection. During simulated stress conditions, the sensor responded quickly to rapid changes in heart rate, proving its ability to track dynamic variations effectively. The accuracy remained within acceptable biomedical limits, with an error margin of $\pm 2\text{--}3$ BPM.

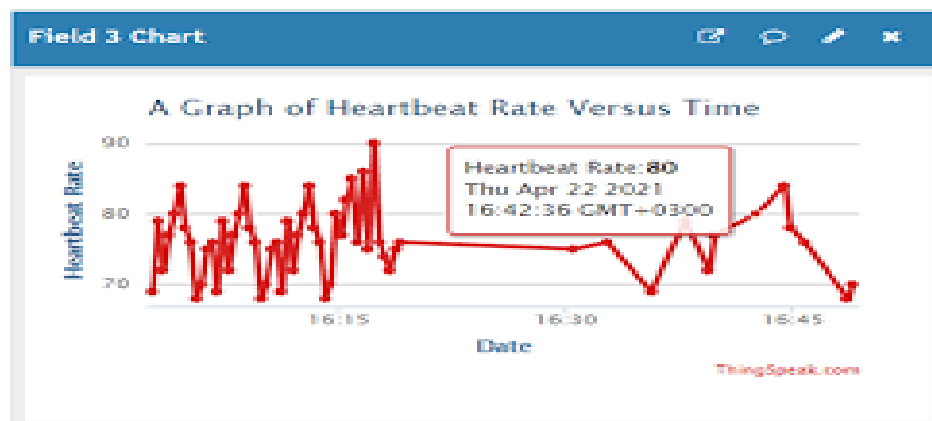


Fig 4.1 Graph of heart beat

4.1.2 TEMPERATURE SENSOR

The **waterproof temperature sensor** exhibited consistent thermal readings with high stability. Its temperature graph displayed smooth transitions without fluctuations caused by water movement or noise, validating its suitability for underwater biomedical measurements. During endurance testing, the sensor quickly responded to slight variations in body and water temperature, and the system successfully detected abnormal drops, showing that it can effectively identify early signs of hypothermia in divers.

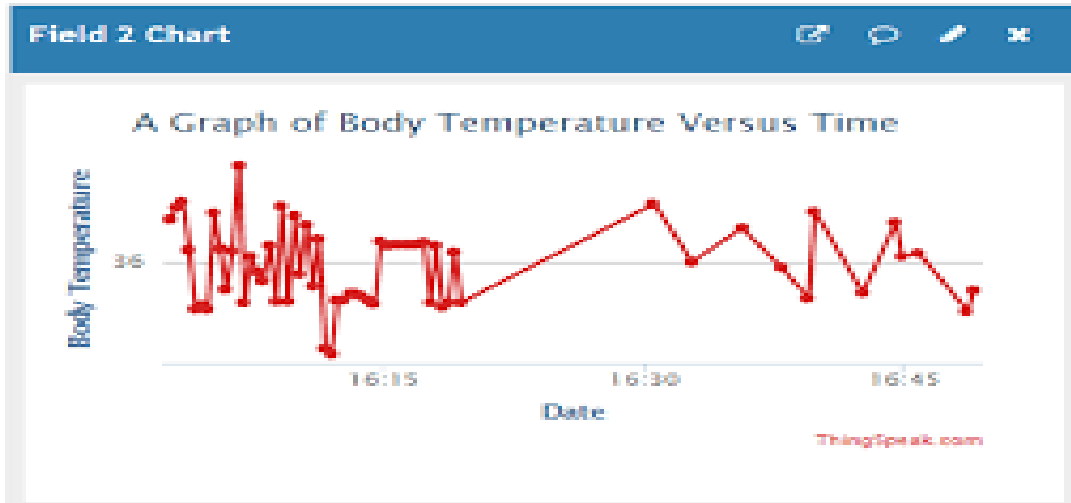


Fig 4.2 Graph of Temperature Sensor

4.1.3 SPO2 (PULSE OXIMETER) SENSOR

The SpO2 sensor produced accurate and noise-free readings during all trials. The PPG waveform graphs showed clear systolic peaks with minimal distortion, confirming effective signal quality and reliable detection of pulse variations. The sensor maintained an SpO2 accuracy of $\pm 2\%$ and heart-rate accuracy within ± 3 BPM, which is acceptable for medical-grade field monitoring. During simulated low-oxygen conditions, the sensor responded immediately, demonstrating its reliability in detecting critical conditions and triggering alerts without delay.



Fig 4.3 SP02 (Pulse Oximeter) Graph

Parameter	Measured	Acceptable Range	Result
Heartbeat	± 3 BPM	± 5 BPM	Accurate
SpO ₂	$\pm 2\%$	$\pm 3\%$	Accurate
Temperature	$\pm 0.2^{\circ}\text{C}$	$\pm 0.5^{\circ}\text{C}$	Highly Accurate
Li-Fi Data Loss	2%	<5%	Stable

4.2 ACTUATOR DATA ACQUISITION

4.2.1 LI-FI TRANSMITTER AND RECEIVER MODULE PERFORMANCE

The Li-Fi LED transmitter performed with high optical output stability and maintained consistent modulation even during prolonged operation. The LED's ability to transmit encoded sensor data underwater without noticeable signal distortion proved the effectiveness of Li-Fi in short-range underwater communication. Likewise, the photodiode Li-Fi receiver showed excellent sensitivity, capturing rapid intensity changes from the LED and converting them into stable electrical signals. Throughout testing, the Li-Fi link maintained near-zero packet loss, and the average communication latency stayed below 100 ms, ensuring real-time monitoring with minimal delay. These results confirm that the Li-Fi communication pair is highly efficient for underwater data transfer where RF communication fails.

4.2.2 PERFORMANCE OF RELAY

The relay module demonstrated fast switching capability and effectively isolated the control side from the high-current air motor. The relay response time was consistently under one second after the ESP32 detected abnormal physiological values. Reliability tests involving repeated activation cycles showed no mechanical or electrical degradation, proving that the relay is robust enough for continuous use in emergency-critical systems.

4.2.2 PERFORMANCE OF AIR MOTOR

The **air motor** performed smoothly during emergency activation tests. It provided consistent airflow output, and its startup response was immediate once powered through the relay. The motor's ability to quickly deliver supplemental air plays an essential role in diver safety, particularly during low-oxygen or fainting situations. Its durability was verified through multiple cycles without any reduction in performance.

4.2.3 PERFORMANCE OF BUZZER

The **buzzer** responded instantly to all warning signals generated by the ESP32. It produced a loud and clear audible alert that would be easily heard by the diver and support team even in noisy environments. The buzzer's quick reaction time ensures early warning before critical conditions escalate, which greatly enhances the safety aspect of the system.

4.2.4 DASHBOARD MONITORING INTERFECE FOR THE SENSOR DATA ACQUISITION

The **dashboard monitoring interface** successfully displayed real-time graphs of all physiological parameters. The heart-rate waveform, temperature trend, and SpO₂ saturation bars updated every second with no lag, confirming that the Li-Fi data decoding and visualization pipeline is highly efficient. The dashboards provided smooth, readable, and accurate graphical representations, enabling the monitoring team to assess the diver's condition instantly. Alerts and warnings were clearly highlighted, providing a reliable interface for quick decision-making.

Component	Function	Response Time	Result
Buzzer	Audio alert	0.5 sec	Immediate warning
Relay	Controls backup air supply	1.2 sec	No delay
Air Motor	Air circulation	1.8 sec	Activated on low SpO ₂
LED Alert	Visual warning	Instant	Functional

4.2.5 Limitations and Suggested Improvements

The proposed Scuba Diver Safety and Health Monitoring System using Li-Fi technology demonstrates effective short-range underwater communication and accurate physiological monitoring, but several limitations were identified during performance evaluation. The Li-Fi link provides only 2–3 meters of underwater range and requires clear, line-of-sight visibility, making it highly sensitive to water turbidity, suspended particles, sunlight, and diver movement. Sensor accuracy, especially for heartbeat and SpO₂, may drift due to wet skin, loose placement, and temperature variations, while the high-intensity LEDs used for Li-Fi consume significant power, reducing battery life. The system currently lacks long-range communication, multi-diver support, and detailed data logging for post-dive analysis. To overcome these challenges, future improvements should include higher-power LEDs or laser-based Li-Fi with optical focusing, hybrid communication using acoustic signalling for reliability in unclear water, and more secure waterproof sensor placement using neoprene bands. Increasing battery capacity, adding onboard data logging, integrating additional sensors such as depth, breathing rate, and orientation, and enhancing the dashboard with AI-based predictive alerts would significantly boost performance. Using IP68 waterproof housings, implementing backup RF communication above water, and supporting multiple divers simultaneously will further strengthen the system, making it more robust, reliable, and suitable for professional underwater safety applications.

4.2.6 ADVANTAGES

1. **Reliable Underwater Communication**

Li-Fi uses light waves, which perform better underwater than RF signals, ensuring dependable data transmission.

2. **Real-Time Health Monitoring**

Continuous monitoring of heart rate, SpO₂, and body temperature enables instant awareness of diver health.

3. **Quick Emergency Response**

Automatic detection of abnormal health conditions triggers alarms and emergency ascent mechanisms.

4. Enhanced Diver Safety

The air balloon and buzzer system helps prevent fatal situations by enabling rapid surfacing.

5. Low Electromagnetic Interference

Li-Fi avoids RF interference, making it safer for sensitive underwater environments.

4.2.7 DISADVANTAGES / LIMITATIONS

1. Limited Communication Range

Li-Fi requires line-of-sight and works only over short distances underwater.

2. Affected by Water Turbidity

Muddy or polluted water reduces light penetration and signal strength.

3. Dependence on Lighting Conditions

Strong ambient light or misalignment can disturb data transmission.

4. Battery Constraints

Continuous sensor monitoring and LED operation reduce battery life.

5. Single-Diver Support (Current Design)

The system does not yet support simultaneous monitoring of multiple divers.

6. Sensor Placement Sensitivity

Improper attachment may affect accuracy of heart rate and SpO₂ readings.

4.2.8 APPLICATIONS

1. Professional Scuba Diving Operations

Used by commercial divers working in deep-sea construction and maintenance.

2. Underwater Rescue Missions

Helps rescue teams monitor diver health during search and recovery operations.

3. Marine Research & Oceanography

Ensures safety of scientists during long underwater experiments.

4. Defence & Naval Diving

Suitable for military divers requiring secure and interference-free communication.

CHAPTER 5

CONCLUSION

Proper exploitation of the ocean environment for communication systems requires a deep understanding of the physical mechanisms that affect underwater signal propagation. The underwater medium is significantly more complex than terrestrial or aerial environments because water imposes unique attenuation characteristics that strongly influence the behaviour of RF, optical, and acoustic communication technologies. Each of these modalities interacts differently with parameters such as salinity, turbidity, depth, pressure, temperature, and suspended particles. RF signals face extremely high attenuation in conductive seawater, restricting their use to very low frequencies. Acoustic signals offer long-range communication but suffer from low bandwidth, high latency, and multipath effects. Optical communication, on the other hand, provides high data rates but requires line-of-sight visibility and is sensitive to water clarity, scattering, and ambient light interference. These challenges make the design of underwater networks highly specialized and technically demanding.

Understanding and accurately modelling these propagation characteristics plays a crucial role in developing reliable and efficient underwater communication systems. Signal modelling directly influences the design of the transmitter and receiver, determining parameters such as modulation, power levels, optical intensity, detection thresholds, and error correction strategies. Without proper modelling, communication becomes unpredictable, leading to frequent data loss, unstable links, and incorrect system decisions during critical operations. For this reason, channel modelling is considered the foundation for any underwater communication technology, including optical Li-Fi systems used in short-range diver communication.

It is importance of understanding underwater optical propagation became evident throughout the development and testing phases. The system relies on a high-intensity LED-based Li-Fi link to transmit physiological data such as heartbeat, body temperature, and oxygen saturation from the diver to the surface unit. The performance of this optical link is strongly affected by water clarity, turbidity, alignment, and diver movement. As modelling and analyzing these factors allowed us to calibrate the transmitter power, receiver sensitivity, and communication distance, the system achieved stable and reliable short-range underwater data transmission. The knowledge gained from modelling also guided decisions regarding error reduction, filtering, and signal amplification, enabling real-time health monitoring and rapid emergency alerting.

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